Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/22/2005
PI Name:	Biswas, Gautam Ph.D.		
Project Title:	Distributed Monitoring and Control of Complex Dynamic Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	37235	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	03/10/2003	**End Date:**	03/09/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kortenkamp, David (Metrica, Inc.)		
Grant/Contract No.:	NCC9-159		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this project is to develop an integrated, robust fault-adaptive control approach for complex dynamic systems by combining distributed monitoring, condition-based maintenance, and decision making to maintain system objectives and support long-term operation of the system. The model-based methodologies developed will be applied to subsystems of the Advanced Life Support System being developed for long-duration manned missions. This proposal falls into the basic research category (levels 1 and 2), but we will evaluate the effectiveness of this methodology by developing advanced control experiments on selected Life Support System problems. The Life Support System for the International Space Station and future manned lunar and Martian exploration provides interesting challenges in that they include a number of interacting control loops, such as the energy management loop, the thermal control loop, the bio-regeneration and gas transfer loop, and the chemical production loop, that cover a variety of domains. An added challenge is that these loops operate at very different time scales, and this will require us to develop		

	innovative abstraction mechanisms and introduce hierarchical model composition techniques to handle concurrency problems in modeling and analysis for autonomous and robust control. Task Objective: Develop multi-level fault-adaptive control schemes to manage complex distributed systems, such as the Advanced Life Support systems for long-duration manned missions. Task Significance: 1. Systematic approaches to modeling complex, distributed systems 2. Distributed algorithms for online monitoring and fault isolation 3. Multi-level fault adaptive controller design that meet resource constraints while optimizing performance over long time horizons
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Water Recovery Systems may be used in desert regions. Advanced autonomous control will make these systems more robust, efficient, and long-lasting. Modeling, monitoring and control technologies and tools developed can be applied to the design and analysis of other complex, dynamic systems.
Task Progress:	This year we extended our work on building dynamic hybrid models of subsystems to integrated models of the Advanced Life Support (ALS) systems. This included Water Recovery, Air Revitalization, Crew Chamber, Power Generation, and the Thermal subsystems. A simulation test bed was created for experimental studies on monitoring and fault diagnosis of ALS subsystems. We recently completed in a NASA JSC run challenge problem on developing life support systems with control to support a 90 day 4 crew manned mission on the lunar surface. The multi-level control scheme developed successfully tied crew schedules and activity to resource managament tasks, took into account the interactions between the subsystems, and optimized the behavior of individual subsystems (ouput versus resources consumed, such as power). The system worked a closed loop system (minimum material vented out) conserving water and air quality to meet the crew activity and EVA schedules. We demonstrated that much tighter bounds on the sizing problem could be attained by considering dynamic system analysis and taking into account the role of sophisticated controllers in the design and analysis process. We have worked closely with two of our collaborators: (i) Dave Kortenkamp and Metrica, Inc. to develop multi-level simulations. In particular, we are linking the BioSim simulation developed at Metrica with our detailed Simulink models of ARS and WRS subsystems. The idea is to provide a comprehensive simulation test bed for running control, diagnosis, scheduling, and planning experiments. (ii) Chris Assad at JPL. We have provided him with Simulink models of our Water recovery system subsystems, and he is developing alternate mechanims for fault detection in these systems.
Bibliography Type:	Description: (Last Updated: 10/05/2020)
Abstracts for Journals and Proceedings	Biswas G, Manders EJ, Ramirez JW. "An Approach to Model Based Diagnosis of ALS subsystems." Habitation 2004: Conference on Space Habitation Research and Technology Development, Orlando, FL, January 2004. Habitation 2004: Conference on Space Habitation Research and Technology Development, Orlando, FL, January 2004. , Jan-2004
Abstracts for Journals and Proceedings	Abdelwahed S, Wu J, Biswas G. "Online Multilevel Hybrid Control for Optimal Resource Management in Advanced Life Support Systems." Habitation 2004: Conference on Space Habitation Research and Technology Development, Orlando, FL, January 2004. Habitation 2004: Conference on Space Habitation Research and Technology Development, Orlando, FL, January 2004. , Jan-2004
Articles in Peer-reviewed Journals	Biswas G, Manders EJ, Ramirez J, Mahadevan N, Abdelwahed S. "Online model-based diagnosis to support autonomous operation of an advanced life support system." Habitation (Elmsford). 2004;10(1):21-37. PMID: 15880907 , Jan-2004
Papers from Meeting Proceedings	Manders EJ, Biswas G, Ramirez JW, Mahadevan N, Wu J. "A Model-Integrated Computing Tool Suite for Fault-Adaptive Control." 15th Annual Workshop on Principles of Diagnosis, Carcassonne, France, June 2004. Proc. 15th Annual Workshop on Principles of Diagnosis, 2004. L. Trave-Massuyes, ed., p. 137-142. , Jun-2004
Papers from Meeting Proceedings	Abdelwahed S, Wu J, Biswas G, Ramirez JW, Manders EJ. "Online Hierarchical Fault-Adaptive Control for Advanced Life-Support Systems." 34th Annual Meeting of Intl. Conf. on Environmental Systems (ICES), Colorado Springs, CO, July 2004. 34th Annual Meeting of Intl. Conf. on Environmental Systems (ICES), Colorado Springs, CO, July 2004. Paper number 2004-01-2441. , Jul-2004